

Tushar Banswal

(+91) 7818035691 | Email Id: tusharbanswal2003@gmail.com | [Linkedin Profile](#)
[Github Profile](#) | [Leetcode Profile](#)

Work Experience

Spraxa Solutions Private Limited | Jan 2024 - Oct 2026 | Noida, Uttar Pradesh, India

Role: Software Engineer

- **Real-Time Telemetry & System:** Engineered low-latency, bidirectional data streaming using MQTT and WebSockets across critical medical equipment tracking modules (EndoScopes, Active Dashboard, Cycle Info).
- **Backend Optimization & Scalability:** Refactored core Node.js/Express REST APIs and restructured MongoDB aggregation pipelines, substantially decreasing query latency (~27%) and system technical debt.
- **Database Architecture & System Integration:** Architected the Scope Compatibility Matrix (SCM) service and integrated it with the CAT platform, normalizing MSSQL relational schema to eliminate state synchronization errors (~35%).
- **Internal Platform Engineering (Dframework):** Co-authored a standardized internal full-stack framework (Node.js/React), integrating enterprise authentication (Entra ID, OAuth, LDAP) and a shared component library that accelerates cross-team feature delivery.
- **CI/CD & Automated Testing:** Built end-to-end automated testing suites using Playwright and TypeScript, integrating them with Azure DevOps and GitHub Actions pipelines to automated test runs and deployment cycles.

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript
- **Backend & Real-time:** Node.js, Express.js, REST APIs, WebSockets, [Socket.io](#), MQTT
- **Database:** MongoDB (Aggregation Pipelines, Mongoose), MSSQL
- **Frontend:** React.js, Next.js, MUI, Svelte, Redux
- **DevOps & QA:** Azure DevOps (CI / CD Pipelines), GitHub Actions, Playwright
- **Architecture & Security:** Multi-Tenant SaaS, RBAC, OAuth/ Entra ID / LDAP.

Projects

- **MyReact — Custom React Implementation** *JavaScript, Babel*
 - Developed a lightweight UI library replicating core React architecture, featuring a custom Virtual DOM tree and reconciliation diffing engine.
 - Implemented custom state lifecycle hooks (useState, useEffect) and integrated custom Babel JSX transpilation to parse nested component hierarchies.
- **t-Lang — Custom Interpreted Language** *JavaScript, Node js, Data Structures.*
 - Design and built an interpreted programming language from scratch, implementing a Lexer for tokenization, an AST generator, and an execution engine.
 - Implemented core language semantics including variable declarations, lexical scoping, arithmetic/logical evaluation, control flow loops, and an interactive CLI REPL.

Education

- **Bachelor of Computer Applications (BCA)** | Anand Engineering College, Agra | 2021 – 2024
 - Score: 82.6%